Continuation Value is All You Need

Preliminary and Incomplete

Jeffrey E. Sun*

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Abstract

I propose a deep-learning method for solving discrete-time heterogeneous-agent models with aggregate shocks by parameterizing the end-of-period continuation value as a function of the beginning-of-period aggregate state for each period. I show that under a set of Lipschitz and contractive conditions, the neural network training residual yields a consistent estimator for the true error between the approximated and true value functions, with explicit bounds.

1 Introduction

Neural Value Function Iteration (nVFI) is an algorithm for solving value functions in heterogeneous-agent models with aggregate shocks. It consists of the following steps:

1. Parameterize the end-of-period value function (a function of both idiosyncratic and aggregate state) using a flexible nonlinear V estimator. (Reference implementation: a neural network.)
2. Draw samples from the aggregate state. (Reference implementation: from the ergodic distribution induced by the current V estimator.)
3. For each aggregate sample, compute the outer Bellman residual—a measure of the failure of the Bellman equation to hold at that aggregate state—using lookahead.
4. Update the V estimator to reduce the outer Bellman residuals across the sampled aggregate states. (Reference implementation: gradient descent on the parameters of the V estimator neural network.)
5. Evaluate global convergence of the V estimator. Return to Step 2 unless the criterion is met.

The algorithm is based on the following insight: if we choose a particular timing convention, then we can define a particular value function object $\mathcal{V}$, the **continuation value operator**. I then show that: if we know $\mathcal{V}$, then we can solve the rest of the model using conventional methods.

This is directly analogous to how, in standard dynamic programming, knowing the value function allows us to solve the model. My view is that $\mathcal{V}$ is the natural generalization of the standard value function to heterogeneous-agent models with aggregate shocks. This is the meaning of the title, "Continuation Value Is All You Need".

*University of Toronto

2 The Heterogeneous-Agent Problem

2.1 Model class and timing conventions

In this section, I specify the class of models for which the algorithm is defined. I specify a timing convention in order to cleanly separate out **aggregate shocks** from the **otherwise-deterministic** evolution of the aggregate state.

The setup is fairly general for discrete-time models, although it may require some translation (without loss of generality) for models written down with different timing conventions.

I also assume throughout that the model is stationary, ergodic, and has a unique global solution.

Time and the household period. Time is discrete and indexed by t. Each period consists of a *household period*, and a moment between household periods. Aggregate shocks are allowed to occur in this moment between household periods.[1]

It may occasionally be useful to refer to intermediate value functions or aggregate states within the period, which will also be denoted with a superscript, e.g. $V^{\text{postincome}}$. These are formally well-defined in terms of stage-decomposed household blocks (see Stages paper).

State. At the start of the period, each household has idiosyncratic state $x^{\text{start}} \in X^{\text{start}}$. At the end of the period, each household has idiosyncratic state $x^{\text{end}} \in X^{\text{end}}$.[2] The aggregate state at the start of period t is Γ_t^{start}; at the end, Γ_t^{end}.

The aggregate state Γ_t^{start} consists of the cross-sectional household state distribution Λ^{start} over x^{start} and other aggregate state variables S^{start}, (and similarly for Γ_t^{end}),

$$\Gamma_t^{\text{start}} = (\Lambda_t^{\text{start}}, S_t^{\text{start}}) \tag{1}$$

$$\Gamma_t^{\text{end}} = (\Lambda_t^{\text{end}}, S_t^{\text{end}}). \tag{2}$$

Aggregate Shocks. An **aggregate shock** is defined as any shock to the model that:

1. Affects the aggregate state of the model, and
2. Is not perfectly foreseen.

Examples include:

- TFP shocks.
- Information revelation about future TFP, in a model with forward-looking behaviour, even if current TFP is unaffected.[3]
- Idiosyncratic productivity shocks to agents who are *granular* as opposed to *atomistic*. That is, shocks to agents whose individual behaviour can affect the aggregate state of the model.

The **only way** aggregate shocks are permitted to enter in this setup is:

1. As unforeseen changes in S_{t+1}^{start} relative to S_t^{end},
2. Occurring between the end of period t and the start of period $t+1$.

[1] Henceforth, I may refer to the beginning (end) of the household period as the "start" ("end") of the period, with the aggregate shock taken to occur "between" periods.

[2] In Krusell–Smith, $x^{\text{start}} = (k^{\text{start}}, z^{\text{start}})$ and $x^{\text{end}} = (k^{\text{end}}, z^{\text{end}})$.

[3] This is an aggregate shock because it may change behaviour in the present.

While this may seem restrictive, it is fairly general:

- If aggregate shocks occur in the middle of the period in a model, since time is periodic, the period boundary can simply be redefined such that the "period" ends immediately before the shock and begins immediately after.
- If information regarding a shock is revealed n periods in advance, this can be formulated as a shock to an information set which is included in the $S^{\cdot}$.
- If shocks "immediately" affecting agents is part of the model, these can be formulated as shocks to the information set contained in S^{start}, which are actually physically resolved only within the period.

I also impose that, for any individual household, $x_t^{\text{end}} = x_{t+1}^{\text{start}}$ through any $t \to t+1$ transition.

Aggregate law of motion between periods. Given these timing assumptions for the aggregate shock(s), we can write the law of motion for S_t without loss of generality as,

$$\Gamma_{t+1}^{\text{start}} = (\Lambda_{t+1}^{\text{start}}, S_{t+1}^{\text{start}}) \tag{3}$$

$$\text{where} \quad \Lambda_{t+1}^{\text{start}} = \Lambda_t^{\text{end}} \tag{4}$$

$$S_{t+1}^{\text{start}} = \Omega(S_t^{\text{end}}, Z_t) \tag{5}$$

$$Z_t \sim \mathcal{D}, \tag{6}$$

where Z_t is the "random part" or innovation in the $S_t \to S_{t+1}$ transition, and $\mathcal{D}$ is its distribution.

In one line,

$$\Gamma_{t+1}^{\text{start}} = (\Lambda_t^{\text{end}}, \Omega(S_t^{\text{end}}, Z_t)), \quad \text{where} \quad Z_t \sim \mathcal{D}. \tag{7}$$

Information revelation. This setup implies the following fact, which we will use: The evolution of the aggregate state *within* the household period, from Γ_t^{start} to Γ_t^{end}, is **deterministic**.

In other words, the aggregate state at the end of the period, Γ_t^{end}, is uniquely determined by the aggregate state at the start of the period, Γ_t^{start}.[4]

2.2 Value functions

We have multiple notions of "the value function" to deal with. In this section I define and disambiguate them.

Full global value function. The full global value function at the start of the period, V^{start}, is a function of both the idiosyncratic state x^{start} and the aggregate state Γ^{start},[5]

$$V^{\text{start}} : (x^{\text{start}}, \Gamma^{\text{start}}) \mapsto V^{\text{start}}(x^{\text{start}}, \Gamma^{\text{start}}) \in \mathbb{R}.$$

[4] The endogenous determination of equilibrium quantities such as prices, quantities, populations, market tightnesses, entry, etc., are simply part of the process by which Γ_t^{end} is determined from Γ_t^{start}. In particular, while Γ_t^{end} may *depend* on prices in the sense of behaviour depending on prices, prices themselves are not a state variable contained in Γ^{start} or Γ^{end}.

[5] Explicitly, the expected NPV of future utility for a household in state x^{start} at the start of the period, given the aggregate state Γ^{start}.

Continuation value. We now break with common convention in an **extremely** useful way.

Recall that Γ^{end} is uniquely determined by Γ^{start} by our construction. Because of this, instead of writing the end-of-period value V^{end} in terms of Γ^{end}, as is common, we can write it in terms of Γ^{start} instead.[6] Absusing notation slightly,[7]

$$V^{\text{end}}(x^{\text{end}}, \Gamma^{\text{start}}) \equiv V^{\text{end (common convention)}}(x^{\text{end}}, \Gamma^{\text{end}}(\Gamma^{\text{start}})). \tag{8}$$

While this may seem misaligned, the story is intuitive. A household who begins the period in state x^{start} can only directly observe x^{start} and Γ^{start}, but must make decisions which depend on x^{end}-contingent values at the **end** of the period. $V^{\text{end}}(x^{\text{end}}, \Gamma^{\text{start}})$ intuitively reflects the act of forecasting end-of-period contingent values from the perspective of the start of period t.[8]

My entire strategy is based on this **continuation value** formulation. The core insight is that, if we have access to $V^{\text{end}}(\cdot, \Gamma^{\text{start}})$, or an approximation thereof, then the rest of the problem can be solved using conventional methods.

At this point, we can introduce the **continuation value operator** $\mathcal{V}$,

$$\mathcal{V}\Gamma^{\text{start}}(x^{\text{end}}) \equiv V^{\text{end}}(x^{\text{end}}, \Gamma^{\text{start}}). \tag{9}$$

My perspective is that this $\mathcal{V}$ is the natural generalization of the standard value function to heterogeneous-agent models with aggregate shocks.

Inner value functions. Note that objects like $\mathcal{V}\Gamma^{\text{start}}$ is a function of x^{end} only.[9] I will call such objects "inner value functions" in order to distinguish them from global value functions which also depend on Γ^{start}.

In the following, I will denote such inner value functions, that is any valuation over the state space X^{end}, by v^{end},[10]

$$v^{\text{end}} : x^{\text{end}} \to \mathbb{R}.$$

It may help to think of v^{end} as a vector over a discretized state space X^{end}. This is how it is often represented on the computer: a vector of values at the gridpoints of x^{end}.

2.3 Continuation value is all you need

In this section I state my central claim: If $\mathcal{V}$ is known, then the rest of the model can be solved with conventional methods.

[6] Why can we do this? V^{end} is obviously uniquely determined by x^{end} and Γ^{end}. But Γ^{end} is uniquely determined by Γ^{start} by construction (see previous section). Therefore, V^{end} is uniquely determined by x^{end} and Γ^{start}.

[7] Explicitly, $V^{\text{end}}(x^{\text{end}}, \Gamma^{\text{start}})$ refers to the NPV of future utility for a household in state x^{end} at the end of the period, given the aggregate state Γ^{start} at the start of the period.

[8] For example, a model in which households make relatively simple (e.g. consumption-savings) decisions might be written as,

$$V^{\text{start}}(x^{\text{start}}, \Gamma^{\text{start}}) = \max_a u(a; x^{\text{start}}) + \beta V^{\text{end}}(x^{\text{end}}(a; x^{\text{start}}), \Gamma^{\text{start}}).$$

[9] In computer science, this is sometimes call a "partially applied function." The input Γ^{start} has been applied, but not the input x^{end}.

[10] This can be any function from the state space x^{end} to the reals. We will typically interpret this object as a guess of the slice of V^{end} at a given Γ^{start}, but note that it **does not make sense** to refer to a "true" value of v^{end} since v^{end} is ultimately still a variable.

2.3.1 Setup and assumptions

An important assumption we need is that aggregate uncertainty is the reason why the model is intractable. That is, in the absence of aggregate uncertainty, the model would be tractable.[11] Explicitly,

Assumption 1. If we shortened the time horizon of the model to one period, with exogenously-given initial aggregate state Γ^{start} and exogenously-given end-of-period x^{end}-contingent payoffs $v^{\text{end}}(x^{\text{end}})$ $\forall x^{\text{end}}$, then we could use conventional methods to find all necessary equilibrium objects (e.g. prices, quantities), obtain $v^{\text{start}}(x^{\text{end}})$ by backwards iteration, and simulate forward to obtain Γ^{end}.

Assumption 1 can be rephrased as assuming that we have access to a **within-period solver** Φ which takes a start-of-period aggregate state Γ^{start} and an end-of-period inner value function v^{end}, and returns an end-of-period aggregate state Γ^{end} and start-of-period inner value function v^{start},

$$\Phi: \ (\Gamma^{\text{start}}, v^{\text{end}}) \ \longmapsto \ (\Gamma^{\text{end}}, v^{\text{start}}).$$

For convenience, let us also define,

$$\Phi_\Gamma(\Gamma^{\text{start}}, v^{\text{end}}) \equiv \Phi(\Gamma^{\text{start}}, v^{\text{end}})_1 = \Gamma^{\text{end}}$$
$$\Phi_V(\Gamma^{\text{start}}, v^{\text{end}}) \equiv \Phi(\Gamma^{\text{start}}, v^{\text{end}})_2 = v^{\text{start}}.$$

2.3.2 Within-period solution given $\mathcal{V}$

Proposition 1. *If we have access to the true $\mathcal{V}$, and Assumption 1 holds, then the model can be solved. In particular, the model can be simulated forward correctly and value functions can be computed for any state Γ^{start}.*

Proof. Given Γ^{start}, we can obtain $V^{\text{end}}(\cdot, \Gamma^{\text{start}}) = \mathcal{V}\Gamma^{\text{start}}$.

The within-period solver yields,

$$\Phi(\Gamma^{\text{start}}, V^{\text{end}}(\cdot, \Gamma^{\text{start}})) = (\Gamma^{\text{end}}, V^{\text{start}}(\cdot, \Gamma^{\text{start}})).$$

Finally, by Equation 7, for any Z, we can simulate forward to the beginning of the following period,

$$\Gamma^{\text{next}} = (\Lambda^{\text{end}}(\Gamma^{\text{end}}), \Omega(S^{\text{end}}(\Gamma^{\text{end}}), Z)).$$

Thus, we are able to simulate the model forward by one full period, and can therefore simulate forward by arbitrarily many periods. □

2.4 Outer Bellman equation and operator

Outer Bellman Equation. A model in our setup can thus be abstractly described by the following four equations,

$$\forall x^{\text{end}}, \Gamma^{\text{start}}, \qquad \mathcal{V}\Gamma^{\text{start}}(x^{\text{end}}) = \mathbb{E}_Z[V^{\text{start}}(x^{\text{end}}, \Gamma^{\text{next}})] \tag{10}$$
$$\text{where} \quad V^{\text{start}}(\cdot, \Gamma^{\text{next}}) \equiv \Phi_V(\Gamma^{\text{next}}, \mathcal{V}\Gamma^{\text{next}}) \tag{11}$$
$$\Gamma^{\text{next}} \equiv \Omega(\Gamma^{\text{end}}, Z) \tag{12}$$
$$\Gamma^{\text{end}} \equiv \Phi_\Gamma(\Gamma^{\text{start}}, \mathcal{V}\Gamma^{\text{start}}). \tag{13}$$

[11] If the model is intractable even in the absence of aggregate uncertainty, then the issue lies elsewhere and the algorithm described here will not be helpful.

Equation 10 can be viewed as the **inner Bellman Equation**. It defines V^{end} as the expectation over realizations of the aggregate shock Z of the start-of-next-period value $V^{\text{start}}(\cdot, \Gamma^{\text{next}})$.

Equation 11 defines start-of-next-period $V^{\text{start}}(\cdot, \Gamma^{\text{next}})$ as the result of obtaining end-of-next-period $V^{\text{end}}(\cdot, \Gamma^{\text{next}})$ via $\mathcal{V}$, then backwards-iterating within-period via Φ_V.

Equation 12 defines Γ^{next} as the result of simulating Γ^{end} through the stochastic transition Ω.

Equation 13 defines Γ^{end} as the result of simulating Γ^{start} forward to the end of the period, given V^{end}.

The four equations taking jointly can be viewed as an **Outer Bellman Equation** implicitly defining the **continuation value operator** $\mathcal{V}$.

Outer Bellman Operator. The left hand side of Equation 10 can be thought of as forecasting the end-of-period-t value function from the beginning of period t via $\mathcal{V}$.

The right hand side of Equation 10, together with equations 11-13, can be thought of as forecasting the end-of-period-t value function from the beginning of period t via forecasting to the end of period $t+1$, then backwards-iterating to the end of period t.

Together, they state that these two operations must give the same result.

Explicitly, we can define an **Outer Bellman Operator**,[12]

$$\begin{aligned}
(\mathcal{L}\mathcal{V})\Gamma^{\text{start}}(x^{\text{end}}) &\equiv \mathbb{E}_Z[V^{\text{start}}(x^{\text{end}}, \Gamma^{\text{next}})] \\
\text{where} \quad V^{\text{start}}(\cdot, \Gamma^{\text{next}}) &\equiv \Phi_V(\Gamma^{\text{next}}, \mathcal{V}\Gamma^{\text{next}}) \\
\Gamma^{\text{next}} &\equiv \Omega(\Gamma^{\text{end}}, Z) \\
\Gamma^{\text{end}} &\equiv \Phi_\Gamma(\Gamma^{\text{start}}, \mathcal{V}\Gamma^{\text{start}}).
\end{aligned}$$

The **Outer Bellman Equation** can then be written,

$$\mathcal{V} = \mathcal{L}\mathcal{V}. \tag{14}$$

2.5 Parameterized continuation values

Value function approximations. The curse of dimensionality bites because it is impractical to represent V^{end} as a vector over an unavoidably large discretized state space $X^{\text{end}} \times \mathbb{R}^{\Gamma^{\text{end}}}$. Our strategy is to use a parameterized function approximator to approximate V^{end} instead. Given a guess of parameters θ, we denote the parameterized approximation by,

$$\widehat{V}_\theta^{\text{end}}(x^{\text{end}}, \Gamma^{\text{start}})$$

and the parameterized approximation of the continuation value operator by $\mathcal{V}_\theta$,

$$\mathcal{V}_\theta\Gamma^{\text{start}}(x^{\text{end}}) \equiv \widehat{V}_\theta^{\text{end}}(x^{\text{end}}, \Gamma^{\text{start}}).$$

In my reference implementation, $\mathcal{V}_\theta$ is a neural network taking a vector Γ^{start} to a vector v^{end}, where the vectors are over discretized state spaces X^{start} and X^{end}.[13]

The Neural Bellman Objective What I call **Neural Value Function Iteration** (nVFI) simply refers to training a neural network to satisfy Equation 14. Explicitly,

$$\min_\theta \mathbb{E}_{\Gamma^{\text{start}}, x^{\text{end}}}\left[(\mathcal{V}_\theta\Gamma^{\text{start}}(x^{\text{end}}) - \mathcal{L}\mathcal{V}_\theta\Gamma^{\text{start}}(x^{\text{end}}))^2\right]. \tag{15}$$

[12] In machine learning, this operation is sometimes called "one-period lookahead."

[13] Note that this is also implicitly done by Krusell and Smith (1998), with the differences being that they use a linear interpolation over a grid of X^{end} and moments of Γ^{start}, and train the parameters θ indirectly via fitting laws of motion for aggregates.

3 The nVFI Method

3.1 The nVFI algorithm

Step 1 — Parameterize $\mathcal{V}$ and guess θ.
Choose a flexible nonlinear estimator $\mathcal{V}_\theta$ that takes an aggregate state Γ^{start} as and returns a vector (or other object) approximating $V^{\text{end}}(\cdot, \Gamma^{\text{start}})$. Initialize θ.

Step 2 — Sample Γ^{start}. (i) Guess Γ^{start}.

(ii) Simulate Γ^{start} forward using the existing $\mathcal{V}_\theta$.

(iii) Sample from the resulting trajectory of Γ^{start}, excluding burn-in.

Step 3 — Compute the outer Bellman residual at each sample.
For each Γ_i^{start} in the sample:

(i) Compute $(\mathcal{L}\mathcal{V}_\theta)\Gamma_i^{\text{start}}$ via lookahead:

(a) $v_i^{\text{end}} \leftarrow \mathcal{V}_\theta(\Gamma_i^{\text{start}})$.

(b) $\Gamma_i^{\text{end}} \leftarrow \Phi_\Gamma(\Gamma_i^{\text{start}}, v_i^{\text{end}})$.

(c) For a sample Z_k in the support of $\mathcal{D}$:

- $\Gamma_{i,Z}^{\text{next}} \leftarrow \Omega(\Gamma_i^{\text{end}}, Z)$.
- $v_{i,Z}^{\text{end}} \leftarrow \mathcal{V}_\theta(\Gamma_{i,Z}^{\text{next}})$.
- $v_{i,Z}^{\text{start}} \leftarrow \Phi_V(\Gamma_{i,Z}^{\text{next}}, v_{i,Z}^{\text{end}})$.

(d) Compute $\widehat{v}_i^{\text{end}} \leftarrow \mathbb{E}_Z[\, v_{i,Z}^{\text{start}}\,]$ as a (possibly probability-weighted) mean of the $v_{i,Z}^{\text{start}}$.

(ii) Compute the gradient of the outer Bellman residual, treating $\widehat{v}_i^{\text{end}}$ as a fixed label:[14]:

$$g_i \;\leftarrow\; \frac{\partial}{\partial\theta}\left\|\mathcal{V}_\theta(\Gamma_i^{\text{start}}) - \widehat{v}_i^{\text{end}}\right\|^2,$$

where $\|\cdot\|^2$ integrates the squared deviation across (a sample of) $x^{\text{end}} \in X^{\text{end}}$.

Step 4 — Update θ.
Take a gradient-descent step,

$$\theta \;\leftarrow\; \theta \;-\; \alpha \cdot \frac{1}{N}\sum_{i=1}^{N} g_i.$$

Step 5 — Evaluate convergence.
Repeat Steps 2–4 until

$$\left\|\mathcal{V}_\theta(\Gamma_i^{\text{start}}) - \widehat{v}_i^{\text{end}}\right\|^2 < \varepsilon$$

4 Results on Local and Global Accuracy

In this section I derive probability bounds for the global accuracy of the nVFI approximator, defined as the ergodic mean distance between the approximated and true value functions with respect to the true ergodic distribution of the model.

[14] That is, not computing gradients through the lookahead step

The proof strategy for the main result will be to construct a ladder of intermediate quantities between the sample statistic (the local residuals evaluated at the sampled states) and the population statistic (the global distance between the nVFI value function estimator and the true value function). We will then construct a sequence of bounds relating each element in the ladder to the last. This will chiefly rely on bounding the rate of change of each element in the ladder with respect to the previous element, then multiplying the maximal rate of change at each step with the distance between the previous element and the "truth."

In order to do this carefully, we must define the objects under study, and define notions of "distance," "maximal rate of change," and "size" for each object. Formally, metrics, Lipschitz seminorms, and norms. This also allows us to formally define the two objects we ultimately wish to relate to each other: the local training residual and the true global error in the approximator.

We then make a set of assumptions necessary for the proof, largely on the Lipschitz constants of various objects. These are analogous to bounded-derivative assumptions, generalized to objects which may not be differentiable.

4.1 Setup

As in the main text, let the *within-period solver* be given by,

$$\Phi : (\Gamma, v^{\text{end}}) \longmapsto (\Gamma^{\text{end}}, v^{\text{end}})$$

where the components of Φ are denoted,

$$\begin{aligned} \Phi_\Gamma : (\Gamma, v^{\text{end}}) &\longmapsto \Gamma^{\text{end}} \\ \text{and} \quad \Phi_V : (\Gamma, v^{\text{end}}) &\longmapsto v^{\text{start}}, \end{aligned}$$

and $\Gamma, \Gamma^{\text{end}} \in \mathbb{\Gamma}$ consist of the household state distribution $\Gamma, \Gamma^{\text{end}} \in \mathbb{\Lambda}$ and other aggregate state variables $S^{\text{start}}, S^{\text{end}} \in \mathbb{S}$:

$$\begin{aligned} \Gamma &\equiv (\Gamma, S^{\text{start}}) \\ \Gamma^{\text{end}} &\equiv (\Gamma^{\text{end}}, S^{\text{end}}). \end{aligned}$$

Value functions $v \in \mathbb{V}$ written in lowercase are inner value functions, or vectors over the household state space X, and population distributions Λ are measures over $\sigma(X)$:

$$\begin{aligned} v^{\text{start}}, v^{\text{end}} &: X \to \mathbb{R} \\ \Lambda &: \sigma(X) \to \mathbb{R} \end{aligned}$$

We further assume that X and $\mathbb{S}$ are compact:

Assumption 2 (Compactness and structure of X and $\mathbb{S}$)**.** X is the product of a compact convex subset X_C of $\mathbb{R}^{\dim(X)}$ and a finite set X_D, representing the continuous and discrete household state variables respectively. $\mathbb{S}$ is a compact subset of $\mathbb{R}^{\dim(S)}$.

4.1.1 Deterministic and Stochastic Laws of Motion

Let

$$Z \sim \mathcal{D}$$

be the aggregate shock. Given Z, denote the law of motion for the aggregate state Γ by,

$$\Gamma^{\text{next}} = \Omega(\Gamma^{\text{end}}, Z).$$

Substituting in the within-period solver allows us to write the law of motion in terms of Γ and v^{end} in place of Γ^{end},

$$\Gamma^{\text{next}}(\Gamma, v^{\text{end}}, Z) \equiv \Omega(\Phi_\Gamma(\Gamma, v^{\text{end}}), Z).$$

Denote an arbitrary continuation operator by $\mathcal{V}$, the true operator $\mathcal{V}^*$, and a parameterized operator $\mathcal{V}_\theta$ with parameters $\theta \in \Theta$. Let the space of continuation operators under consideration be denoted,

$$\mathcal{N} \equiv \{\mathcal{V}^\star\} \cup \{\mathcal{V}_\theta : \theta \in \Theta\}$$

Any $\mathcal{V}$ induces a law of motion for Γ which we denote,

$$\Gamma^{\text{next}}_{\mathcal{V}}(\Gamma, Z) \equiv \Gamma^{\text{next}}(\Gamma, \mathcal{V}\Gamma, Z).$$

4.1.2 Metrics and Norms

Metrics. C and $\mathbb{S}$ take the usual Euclidean norm, denoted $|\cdot|$ throughout. The metric on household states $x = (x_c, x_d)$ is,

$$d\left(x, \tilde{x}\right) \equiv |x_c - \tilde{x}_c| + 2\,\mathbf{1}[x_d \neq \tilde{x}_d].$$

For distributions, such as the household state distribution[15] Λ, define the Wasserstein-1 metric,

$$W_1(\Lambda, \widetilde{\Lambda}) \equiv \inf_{\pi \in \mathcal{C}(\Lambda, \widetilde{\Lambda})} \int_{X \times X} d\left(x, \tilde{x}\right)\, \mathrm{d}\pi(x, \tilde{x}),$$

representing the minimal total distance required to move Λ onto $\widetilde{\Lambda}$ by moving $\pi(x, \tilde{x})$ households from each idiosyncratic state x to $\tilde{x}$.

The aggregate state space $\mathbb{\Gamma}$ takes the metric,

$$d\left(\Gamma, \widetilde{\Gamma}\right) \equiv W_1(\Lambda, \widetilde{\Lambda}) + |S - \widetilde{S}|.$$

Define the *diameter* $\Delta_{\mathbb{\Gamma}}$ as the maximal distance between any two points in $\mathbb{\Gamma}$,

$$\Delta_{\mathbb{\Gamma}} = \sup_{\Gamma, \widetilde{\Gamma} \in \mathbb{\Gamma}} d\left(\Gamma, \widetilde{\Gamma}\right),$$

with $\Delta_{\mathbb{\Gamma}} < \infty$ by compactness of $\mathbb{\Gamma}$.

Lemma 1. *$\mathbb{\Gamma}$ is compact.*

Proof. By Assumption 2, X is compact. By (**?**, Remark 6.19), it is a property of W_1 that the set $\mathcal{P}(X)$ of probability measures on a compact set X is compact under W_1. Assumption 2 also states that $\mathbb{S}$ is compact. The product of compact metric spaces is compact under the product measure, and the compactness of $\mathbb{\Gamma} = \mathcal{P}(X) \times \mathbb{S}$ under $d\left(\cdot, \cdot\right)$ defined above follows from this. □

Lipschitz Seminorms. The Lipschitz seminorm of an function f, roughly, captures the supremum of its local slopes and is defined even when the object is not differentiable. Formally,

$$\text{Lip}(f) = \sup_{x \neq \tilde{x}} \frac{d\left(f(x), f(\tilde{x})\right)}{d\left(x, \tilde{x}\right)}.$$

In the case f is an inner value function with domain X, we define the Lipschitz seminorm Lip_c which restricts to continuous changes,

$$\text{Lip}_c(v) = \sup_{x \neq \tilde{x} | x_d = \tilde{x}_d} \frac{d\left(v(x), v(\tilde{x})\right)}{|x_c - \tilde{x}_c|}.$$

[15] Later on, the Wasserstein-1 metric will also be used for distributions over aggregate states.

Norms. The inner value function takes the norm,

$$\|v\| \equiv \max\left\{\sup_x |v(x)|,\ \mathrm{Lip}(v)\right\}.$$

For an operator $\mathcal{V}$ on Γ, define the norm and Lipschitz seminorm,

$$\|\mathcal{V}\| \equiv \sup_\Gamma \|\mathcal{V}\Gamma\|, \qquad \mathrm{Lip}(\mathcal{V}) = \sup_{\Gamma\neq\widetilde{\Gamma}} \frac{\left\|\mathcal{V}\Gamma - \mathcal{V}\widetilde{\Gamma}\right\|}{d\left(\Gamma,\widetilde{\Gamma}\right)}.$$

For a function f on Γ, define the norms,

$$\|f\| \equiv \sup_\Gamma |f|,$$

which is defined without reference to a measure on $\mathbb{\Gamma}$, and

$$\|f\|_{p,\psi} \equiv \left(\int |f|^p \,\mathrm{d}\psi\right)^{1/p},$$

for any measure ψ over $\mathbb{\Gamma}$ and $p > 0$.

Residual and error. Let $\mathcal{V}_\theta$ be an approximator and $\mathcal{V}^*$ be the true value function operator on Γ. The main result we are building toward is relating the residuals computed locally for aggregate states drawn from a sampling distribution to the true error between $\mathcal{V}_\theta$ and $\mathcal{V}^*$. We can now formally define these quantities.

The *local residual* at an aggregate state Γ is the norm of the error in the Outer Bellman Equation, by lookahead under the approximator $\mathcal{V}_\theta$,

$$\begin{aligned} r_\theta(\Gamma) &\equiv \|\mathcal{V}_\theta\Gamma - (\mathcal{L}\mathcal{V}_\theta)\Gamma\| \\ \text{with}\quad r_\theta^2(\Gamma) &\equiv r_\theta(\Gamma)^2, \end{aligned}$$

where the norm is the norm on inner value functions v defined above.

The *local true error* is the norm of the difference between the inner value function under $\mathcal{V}_\theta$ and the true one under $\mathcal{V}^*$,

$$\varepsilon_\theta(\Gamma) \equiv \|\mathcal{V}_\theta\Gamma - \mathcal{V}^*\Gamma\|.$$

The ergodic mean error is a global quantity, defined as the integral of the local true error,

$$\|\varepsilon_\theta\|_{p,\psi_{\mathcal{V}^*}} \equiv \left(\int \varepsilon_\theta^p \,\mathrm{d}\psi_{\mathcal{V}^*}\right)^{1/p},$$

for $p > 0$ and $\psi_{\mathcal{V}^*}$ the true ergodic distribution of Γ. The main result concerns the cases $p = 1$ and $p = 2$. The formal definition of the ergodic distribution, and conditions for its existence, are deferred to Section X.

4.2 Assumptions

The first three assumptions state that the influence of the aggregate state Γ on the (true) continuation value v^{end}, and the influence of the continuation value the beginning-of-period value v^{start} and the beginning-of-next-period state Γ^{next}, respectively, are all bounded. The fourth assumption Φ_V places a joint restriction on those bounds.

Assumption 3 (True continuation value Lipschitz in the aggregate state)**.** There is a constant $L_V < \infty$ such that for all $\Gamma, \widetilde{\Gamma} \in \mathbb{\Gamma}$,

$$\begin{aligned} \left\| v^{\text{start}} - \tilde{v}^{\text{start}} \right\| &\leq L_V \, d(\Gamma, \widetilde{\Gamma}), \\ \text{where} \quad v^{\text{start}} &= \Phi_V(\Gamma, \mathcal{V}^\star \Gamma), \\ \tilde{v}^{\text{start}} &= \Phi_V(\widetilde{\Gamma}, \mathcal{V}^\star \widetilde{\Gamma}) \end{aligned}$$

Assumption 4 (Φ_V Lipschitz in the continuation value)**.** For any $\Gamma \in \mathbb{\Gamma}$, there exists a finite *local contraction modulus* $\xi(\Gamma)$ of Φ_V at Γ, defined as

$$\begin{aligned} \xi(\Gamma) = \sup_{v^{\text{end}} \neq \tilde{v}^{\text{end}}} & \frac{\left\| v^{\text{start}} - \tilde{v}^{\text{start}} \right\|}{\left\| v^{\text{end}} - \tilde{v}^{\text{end}} \right\|}, \\ \text{s.t.} \quad v^{\text{start}} &= \Phi_V(\Gamma, v^{\text{end}}), \\ \tilde{v}^{\text{start}} &= \Phi_V(\Gamma, \tilde{v}^{\text{end}}) \end{aligned}$$

Furthermore, these local contraction moduli are bounded above by a finite constant,

$$\xi_{\max} = \sup_{\Gamma \in \mathbb{\Gamma}} \xi(\Gamma) < \infty.$$

Assumption 5 (Law of motion Lipschitz in the continuation value)**.** There exists $L_G < \infty$ such that for all $v^{\text{end}}, \tilde{v}^{\text{end}}$, all $\Gamma \in \mathbb{\Gamma}$, and all Z,

$$\begin{aligned} d\big(\Gamma^{\text{next}}, \widetilde{\Gamma}^{\text{next}}\big) &\leq L_G \left\| v^{\text{end}} - \tilde{v}^{\text{end}} \right\|, \\ \text{where} \quad \Gamma^{\text{next}} &= \Gamma^{\text{next}}(\Gamma, v^{\text{end}}, Z), \\ \widetilde{\Gamma}^{\text{next}} &= \Gamma^{\text{next}}(\Gamma, \tilde{v}^{\text{end}}, Z). \end{aligned}$$

Assumption 6 (Joint contractionary restriction)**.** $\xi_{\max} + L_V L_G < 1$.[16]

Intuitively, Assumption 6 ensures that the usual contraction modulus $\xi_{\max}$ is not overwhelmed by the general-equilibrium effect of v^{end} on itself through its effect on the law of motion Γ cascading forward to $v^{\text{end,next}}$ and then back to v^{end}.

The next two assumptions state that, holding v^{end} constant, we can bound the influence of the aggregate state Γ on the current value v^{start} and the beginning-of-next-period state Γ^{next}, respectively. The following assumption states that the influence of Γ on v^{end} itself is bounded, globally across all operators $\mathcal{V}$ under consideration.

[16] The sharp restriction, unlikely to differ much is practice, is, for all $\mathcal{V} \in \mathcal{N}$ and $\Gamma \in \mathbb{\Gamma}$,

$$\sqrt{\mathbb{E}_Z\left[\xi\big(\Gamma^{\text{next}}_{\mathcal{V}}(\Gamma, Z)\big)^2\right]} + L_V L_G < 1.$$

Assumption 7 (Law of motion Lipschitz in the aggregate state)**.** There is a constant $L_{G,\Gamma} < \infty$ such that for all v^{end}, all $\Gamma, \widetilde{\Gamma} \in \mathbb{\Gamma}$, and all Z,

$$\begin{aligned} d\big(\Gamma^{\text{next}}, \widetilde{\Gamma}^{\text{next}}\big) &\leq L_{G,\Gamma}\, d(\Gamma, \widetilde{\Gamma}) \\ \text{where} \quad \Gamma^{\text{next}} &= \Gamma^{\text{next}}(\Gamma, v^{\text{end}}, Z), \\ \widetilde{\Gamma}^{\text{next}} &= \Gamma^{\text{next}}(\widetilde{\Gamma}, v^{\text{end}}, Z). \end{aligned}$$

Assumption 8 (Φ_V Lipschitz in the aggregate state)**.** There is a constant $L_\Phi < \infty$ such that for all v^{end} and all $\Gamma, \widetilde{\Gamma} \in \mathbb{\Gamma}$,

$$\begin{aligned} \left\| v^{\text{start}} - \tilde{v}^{\text{start}} \right\| &\leq L_\Phi d(\Gamma, \widetilde{\Gamma}) \\ \text{where} \quad v^{\text{start}} &= \Phi_V(\Gamma, v^{\text{end}}), \\ \tilde{v}^{\text{start}} &= \Phi_V(\widetilde{\Gamma}, v^{\text{end}}). \end{aligned}$$

Assumption 9 (True and approximated continuation values Lipschitz in the aggregate state)**.** There is a constant $L_{\mathcal{N}} < \infty$ such that for any $\mathcal{V} \in \mathcal{N}$, and any $\Gamma, \widetilde{\Gamma} \in \mathbb{\Gamma}$,

$$\left\| \mathcal{V}\Gamma - \mathcal{V}\widetilde{\Gamma} \right\| \leq L_{\mathcal{N}}\, d(\Gamma, \widetilde{\Gamma}).$$

Assumption 9 will hold, for example, whenever Θ are the weights and biases of a class of feedforward neural networks with Lipschitz activation functions.

Finally, we also require a relatively standard mixing condition on the law of motion, globally across all continuation value operators $\mathcal{V}$ under consideration.

Henceforth, let $\text{Dist}(X \mid Y)$ denote the conditional distribution of X given Y.

Assumption 10 (Mixing)**.** There is $\rho \in [0, 1)$ such that for every $\mathcal{V} \in \mathcal{N}$ and all $\Gamma, \widetilde{\Gamma}$,

$$\begin{aligned} W_1\left(\text{Dist}\left(\Gamma^{\text{next}} \mid \Gamma\right), \text{Dist}\left(\widetilde{\Gamma}^{\text{next}} \mid \widetilde{\Gamma}\right)\right) &\leq \rho\, d(\Gamma, \widetilde{\Gamma}), \\ \text{where} \quad \Gamma^{\text{next}} &= \Gamma^{\text{next}}_{\mathcal{V}}(\Gamma, Z), \\ \widetilde{\Gamma}^{\text{next}} &= \Gamma^{\text{next}}_{\mathcal{V}}(\widetilde{\Gamma}, Z). \end{aligned}$$

4.3 Local Inequalities

Before moving on to the key chain of inequalities (Lemmas X, X, and X), we first must set up some lemmas concerning local bounds and global ergodicity. The key chain of inequalities will then use these to prove statements about the ergodic means of local residuals and errors, that is, about global residuals and errors.

The first step is to show a bound on the local error, which comes with a nuisance term that will disappear in the ergodic case. Corollary 1 makes the connection to Assumption 6 explicit. Then we will show that the local residual is itself Lipschitz with an explicit upper bound on its Lipschitz constant. The last piece of setup will be to define and prove the existence of ergodic distributions for the laws of motion induced by $\mathcal{V} \in \mathcal{N}$.

Lemma 2 (Local error is bounded by local residual and contracted next-period error)**.** *Under Assumptions 3–5, for every $\theta \in \Theta$ and every $\Gamma \in \mathbb{\Gamma}$, the* local residual r_θ *and the* local true error ε_θ *satisfy*

$$\begin{aligned} (1 - L_V L_G)\, \varepsilon_\theta(\Gamma) - \mathbb{E}_Z\left[\xi(\Gamma^{\text{next}}_\theta)\, \varepsilon_\theta(\Gamma^{\text{next}}_\theta) \;\middle|\; \Gamma\right] &\leq r_\theta(\Gamma), \qquad (16) \\ \textit{where} \quad \Gamma^{\text{next}}_\theta &\equiv \Gamma^{\text{next}}_{\mathcal{V}_\theta}(\Gamma, Z). \end{aligned}$$

Proof. Fix Γ and Z, and let $\Gamma_\star^{\text{next}} \equiv \Gamma_{\mathcal{V}^\star}^{\text{next}}(\Gamma, Z)$ be next period's aggregate state under $\mathcal{V}^\star$. Since $\mathcal{V}^\star = \mathcal{L}\mathcal{V}^\star$,

$$\begin{aligned}\varepsilon_\theta(\Gamma) &\equiv \|\mathcal{V}_\theta\Gamma - \mathcal{V}^\star\Gamma\| \\ &= \big\|\big[\mathcal{V}_\theta\Gamma - (\mathcal{L}\mathcal{V}_\theta)\Gamma\big] + \big[(\mathcal{L}\mathcal{V}_\theta)\Gamma - (\mathcal{L}\mathcal{V}^\star)\Gamma\big]\big\|\end{aligned}$$

By the definition of $\mathcal{L}$,

$$\begin{aligned}(\mathcal{L}\mathcal{V}_\theta)\Gamma - (\mathcal{L}\mathcal{V}^\star)\Gamma = \mathbb{E}_Z\big[\Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}_\theta\Gamma_\theta^{\text{next}}) - \Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}^\star\Gamma_\theta^{\text{next}}) \;\big|\; \Gamma\big] \\ + \mathbb{E}_Z\big[\Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}^\star\Gamma_\theta^{\text{next}}) - \Phi_V(\Gamma_\star^{\text{next}}, \mathcal{V}^\star\Gamma_\star^{\text{next}}) \;\big|\; \Gamma\big].\end{aligned}$$

By the triangle inequality,

$$\begin{aligned}\varepsilon_\theta(\Gamma) \le \|\mathcal{V}_\theta\Gamma - (\mathcal{L}\mathcal{V}_\theta)\Gamma\| \\ + \big\|\mathbb{E}_Z\big[\Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}_\theta\Gamma_\theta^{\text{next}}) - \Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}^\star\Gamma_\theta^{\text{next}}) \;\big|\; \Gamma\big]\big\| \\ + \big\|\mathbb{E}_Z\big[\Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}^\star\Gamma_\theta^{\text{next}}) - \Phi_V(\Gamma_\star^{\text{next}}, \mathcal{V}^\star\Gamma_\star^{\text{next}}) \;\big|\; \Gamma\big]\big\|.\end{aligned}$$

First,

$$\|\mathcal{V}_\theta\Gamma - (\mathcal{L}\mathcal{V}_\theta)\Gamma\| = r_\theta(\Gamma).$$

Next, since $\|\cdot\|$ is a supremum of absolute values of linear functionals, $\|\mathbb{E}_Z F\| \le \mathbb{E}_Z \|F\|$ for any F. In particular,

$$\begin{aligned}&\big\|\mathbb{E}_Z\big[\Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}_\theta\Gamma_\theta^{\text{next}}) - \Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}^\star\Gamma_\theta^{\text{next}}) \;\big|\; \Gamma\big]\big\| \\ &\le \mathbb{E}_Z\big[\big\|\Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}_\theta\Gamma_\theta^{\text{next}}) - \Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}^\star\Gamma_\theta^{\text{next}})\big\| \;\big|\; \Gamma\big] \\ &\le \mathbb{E}_Z\big[\xi(\Gamma_\theta^{\text{next}})\,\varepsilon_\theta(\Gamma_\theta^{\text{next}}) \mid \Gamma\big], \quad \text{by Assumption 4.}\end{aligned}$$

Similarly, by Assumption 3 and then Assumption 5,

$$\begin{aligned}\big\|\Phi_V(\Gamma_\theta^{\text{next}}, \mathcal{V}^\star\Gamma_\theta^{\text{next}}) - \Phi_V(\Gamma_\star^{\text{next}}, \mathcal{V}^\star\Gamma_\star^{\text{next}})\big\| &\le L_V\, d(\Gamma_\theta^{\text{next}}, \Gamma_\star^{\text{next}}) \\ &\le L_V L_G\, \|\mathcal{V}_\theta\Gamma - \mathcal{V}^\star\Gamma\| \\ &= L_V L_G\, \varepsilon_\theta(\Gamma),\end{aligned}$$

So that all together,

$$\varepsilon_\theta(\Gamma) \le r_\theta(\Gamma) + \mathbb{E}_Z\big[\xi(\Gamma_\theta^{\text{next}})\,\varepsilon_\theta(\Gamma_\theta^{\text{next}}) \;\big|\; \Gamma\big] + L_V L_G\, \varepsilon_\theta(\Gamma),$$

yielding (16) by rearrangement. □

Corollary 1 (The local error is bounded by the local residual)**.** *Under Assumption 6,*

$$\varepsilon_\theta(\Gamma) \le \frac{r_\theta(\Gamma)}{1 - \xi_{\max} - L_V L_G} + \frac{\mathbb{E}_Z\left[\xi\left(\Gamma_\theta^{\text{next}}\right)\varepsilon_\theta(\Gamma_\theta^{\text{next}}) \mid \Gamma\right] - \xi_{\max}\varepsilon_\theta(\Gamma)}{1 - \xi_{\max} - L_V L_G}$$

$$\textit{where} \quad \Gamma_\theta^{\text{next}} = \Gamma_{\mathcal{V}_\theta}^{\text{next}}(\Gamma, Z).$$

This almost gives us what we want—a bound on the local true error in terms of the local residual—but it comes with a nuisance term in terms of the expected error in the next state. Lemma 6 will show that this nuisance term vanishes when we take ergodic means.

Lemma 3 (The local residual and loss are Lipschitz)**.** *Under Assumptions 4, 5 and 7–9, for any* $\theta \in \Theta$ *and any* $\Gamma, \widetilde{\Gamma} \in \mathbb{\Gamma}$*,*

$$\begin{aligned}\left|r_\theta(\Gamma) - r_\theta(\widetilde{\Gamma})\right| &\le L_r\, d(\Gamma, \widetilde{\Gamma}) \\ \left|r_\theta^2(\Gamma) - r_\theta^2(\widetilde{\Gamma})\right| &\le 2\,\|r_\theta\|\, L_r\, d(\Gamma, \widetilde{\Gamma})\end{aligned}$$

$$\textit{where} \quad L_r \equiv L_{\mathcal{N}} + (L_\Phi + \xi_{\max} L_{\mathcal{N}})(L_{G,\Gamma} + L_G L_{\mathcal{N}}).$$

Proof. Since $r_\theta(\Gamma) \equiv \|\mathcal{V}_\theta\Gamma - (\mathcal{L}\mathcal{V}_\theta)\Gamma\|$, it immediately follows that,

$$\begin{aligned}\mathrm{Lip}(r_\theta) &\le \mathrm{Lip}(\mathcal{V}_\theta) + \mathrm{Lip}(\mathcal{L}\mathcal{V}_\theta) \\ &\le L_{\mathcal{N}} + \mathrm{Lip}(\mathcal{L}\mathcal{V}_\theta) \quad \text{by Assumption 9.}\end{aligned}$$

It remains to bound $\mathrm{Lip}(\mathcal{L}\mathcal{V}_\theta)$. Fix $\Gamma, \widetilde{\Gamma}$ and Z, and let,

$$\begin{aligned}\Gamma^{\mathrm{next}} &\equiv \Gamma^{\mathrm{next}}_{\mathcal{V}_\theta}(\Gamma, Z) \\ \widetilde{\Gamma}^{\mathrm{next}} &\equiv \Gamma^{\mathrm{next}}_{\mathcal{V}_\theta}(\widetilde{\Gamma}, Z).\end{aligned}$$

By the triangle inequality,

$$\begin{aligned}&\left\|\Phi_V(\Gamma^{\mathrm{next}}, \mathcal{V}_\theta\Gamma^{\mathrm{next}}) - \Phi_V(\widetilde{\Gamma}^{\mathrm{next}}, \mathcal{V}_\theta\widetilde{\Gamma}^{\mathrm{next}})\right\| \\ &\le \left\|\Phi_V(\Gamma^{\mathrm{next}}, \mathcal{V}_\theta\Gamma^{\mathrm{next}}) - \Phi_V(\widetilde{\Gamma}^{\mathrm{next}}, \mathcal{V}_\theta\Gamma^{\mathrm{next}})\right\| + \left\|\Phi_V(\widetilde{\Gamma}^{\mathrm{next}}, \mathcal{V}_\theta\Gamma^{\mathrm{next}}) - \Phi_V(\widetilde{\Gamma}^{\mathrm{next}}, \mathcal{V}_\theta\widetilde{\Gamma}^{\mathrm{next}})\right\| \\ &\le \left(L_\Phi + \xi(\widetilde{\Gamma}^{\mathrm{next}})\, L_{\mathcal{N}}\right) d(\Gamma^{\mathrm{next}}, \widetilde{\Gamma}^{\mathrm{next}}) \quad \text{by Assumptions 8, 4 and 9,}\end{aligned}$$

and similarly,

$$\begin{aligned}d(\Gamma^{\mathrm{next}}, \widetilde{\Gamma}^{\mathrm{next}}) &\le d\big(\Gamma^{\mathrm{next}}(\Gamma, \mathcal{V}_\theta\Gamma, Z), \Gamma^{\mathrm{next}}(\widetilde{\Gamma}, \mathcal{V}_\theta\Gamma, Z)\big) + d\big(\Gamma^{\mathrm{next}}(\widetilde{\Gamma}, \mathcal{V}_\theta\Gamma, Z), \Gamma^{\mathrm{next}}(\widetilde{\Gamma}, \mathcal{V}_\theta\widetilde{\Gamma}, Z)\big) \\ &\le (L_{G,\Gamma} + L_G L_{\mathcal{N}})\, d(\Gamma, \widetilde{\Gamma}) \quad \text{by Assumptions 7, 5 and 9.}\end{aligned}$$

Hence,

$$\begin{aligned}&\left\|\Phi_V(\Gamma^{\mathrm{next}}, \mathcal{V}_\theta\Gamma^{\mathrm{next}}) - \Phi_V(\widetilde{\Gamma}^{\mathrm{next}}, \mathcal{V}_\theta\widetilde{\Gamma}^{\mathrm{next}})\right\| \\ &\le \left(L_\Phi + \xi(\widetilde{\Gamma}^{\mathrm{next}})\, L_{\mathcal{N}}\right) d(\Gamma^{\mathrm{next}}, \widetilde{\Gamma}^{\mathrm{next}}), \\ &\le \left(L_\Phi + \xi(\widetilde{\Gamma}^{\mathrm{next}})\, L_{\mathcal{N}}\right)(L_{G,\Gamma} + L_G L_{\mathcal{N}})\, d(\Gamma, \widetilde{\Gamma}) \quad \text{by Assumptions 7, 5 and 9} \\ &\le \left(L_\Phi + \xi_{\max} L_{\mathcal{N}}\right)(L_{G,\Gamma} + L_G L_{\mathcal{N}})\, d(\Gamma, \widetilde{\Gamma})\end{aligned}$$

By the definition of $\mathcal{L}$, and since $\|\mathbb{E}_Z F\| \le \mathbb{E}_Z \|F\|$ for any F,

$$\begin{aligned}\left\|(\mathcal{L}\mathcal{V}_\theta)\Gamma - (\mathcal{L}\mathcal{V}_\theta)\widetilde{\Gamma}\right\| &= \left\|\mathbb{E}_Z\left[\Phi_V(\Gamma^{\mathrm{next}}, \mathcal{V}_\theta\Gamma^{\mathrm{next}}) - \Phi_V(\widetilde{\Gamma}^{\mathrm{next}}, \mathcal{V}_\theta\widetilde{\Gamma}^{\mathrm{next}})\right]\right\| \\ &\le \mathbb{E}_Z\left\|\Phi_V(\Gamma^{\mathrm{next}}, \mathcal{V}_\theta\Gamma^{\mathrm{next}}) - \Phi_V(\widetilde{\Gamma}^{\mathrm{next}}, \mathcal{V}_\theta\widetilde{\Gamma}^{\mathrm{next}})\right\| \\ &\le (L_\Phi + \xi_{\max} L_{\mathcal{N}})(L_{G,\Gamma} + L_G L_{\mathcal{N}})\, d(\Gamma, \widetilde{\Gamma}).\end{aligned}$$

Therefore,

$$\mathrm{Lip}(r_\theta) \le L_{\mathcal{N}} + (L_\Phi + \xi_{\max} L_{\mathcal{N}})(L_{G,\Gamma} + L_G L_{\mathcal{N}}) \equiv L_r.$$

Finally,

$$\begin{aligned}\left|r_\theta^2(\Gamma) - r_\theta^2(\widetilde{\Gamma})\right| &= \left|r_\theta(\Gamma) - r_\theta(\widetilde{\Gamma})\right| \left(r_\theta(\Gamma) + r_\theta(\widetilde{\Gamma})\right) \\ &\le L_r\, d(\Gamma, \widetilde{\Gamma}) \cdot 2\,\|r_\theta\|\,.\end{aligned}$$

□

4.4 Ergodicity and regularity

Definition 1 (Invariant and Ergodic Distribution)**.** Let $\mathcal{V} \in \mathcal{N}$ be an operator. Consider a trajectory $\{\Gamma_t\}_{t\geq 0}$ under $\mathcal{V}$,

$$\Gamma_{t+1} = \Gamma_{\mathcal{V}}^{\text{next}}(\Gamma_t, Z_t),$$
$$Z_t \sim \mathcal{D} \text{ i.i.d.}$$

An *invariant distribution* for $\mathcal{V}$ is a distribution ψ on $\mathbb{\Gamma}$ for which,

$$\text{Dist}(\Gamma_1 \mid \Gamma_0 \sim \psi) = \psi.$$

If ψ is the *unique* invariant distribution for $\mathcal{V}$, then $\psi_{\mathcal{V}} \equiv \psi$ is its ergodic distribution. In particular,

$$\psi_\theta \equiv \psi_{\mathcal{V}_\theta}$$
$$\psi^\star \equiv \psi_{\mathcal{V}^\star},$$

denote the ergodic distributions of $\mathcal{V}_\theta$ and $\mathcal{V}^\star$. The following lemma proves they exist.

Lemma 4 (Simulation under $\mathcal{V} \in \mathcal{N}$ contracts the space of distributions on $\mathbb{\Gamma}$.)**.** *Under Assumptions 2 and 10, every $\mathcal{V} \in \mathcal{N}$ has a unique ergodic distribution $\psi_{\mathcal{V}}$. Furthermore, for any distributions $\psi, \tilde{\psi}$ on $\mathbb{\Gamma}$ and any $t \geq 0$, the trajectories under $\mathcal{V}$ satisfy,*

$$W_1\big(\text{Dist}(\Gamma_t \mid \Gamma_0 \sim \psi), \text{Dist}(\Gamma_t \mid \Gamma_0 \sim \tilde{\psi})\big) \leq \rho^t\, W_1(\psi, \tilde{\psi}),$$

where ρ is the mixing constant of Assumption 10.

Proof. Let,

$$\Gamma_1 \equiv \Gamma_{\mathcal{V}}^{\text{next}}(\Gamma_0, Z_0) \qquad \widetilde{\Gamma}_1 \equiv \Gamma_{\mathcal{V}}^{\text{next}}(\widetilde{\Gamma}_0, Z_0)$$
$$\Gamma_0 \sim \psi \qquad \widetilde{\Gamma}_0 \sim \tilde{\psi}$$

Under the mixing condition of Assumption 10, Kantorovich–Rubinstein duality (Lemma 8) states that for any 1-Lipschitz f on $\mathbb{\Gamma}$, the function $f_1(\Gamma) \equiv \mathbb{E}_Z f\big(\Gamma_{\mathcal{V}}^{\text{next}}(\Gamma, Z)\big)$ is ρ-Lipschitz. That is,

$$\mathbb{E}f(\Gamma_1) - \mathbb{E}f(\widetilde{\Gamma}_1) = \int f_1 \,\mathrm{d}\psi - \int f_1 \,\mathrm{d}\tilde{\psi} \leq \rho\, W_1(\psi, \tilde{\psi}),$$

and the supremum over f gives the case $t = 1$. The case t follows by induction.

The set of probability measures on the compact $\mathbb{\Gamma}$ is compact under W_1 (**?**, Remark 6.19) and therefore complete. By the Banach fixed-point theorem, because the map sending ψ to $\text{Dist}(\Gamma_1)$ is a ρ-contraction, it has a unique fixed point. This fixed point is the unique invariant distribution $\psi_{\mathcal{V}}$. □

Corollary 2 (Every distribution contracts towards the ergodic distribution)**.** *Under Assumptions 2 and 10, for every $\mathcal{V} \in \mathcal{N}$, any distribution ψ on $\mathbb{\Gamma}$ contracts towards $\psi_{\mathcal{V}}$ at rate at least ρ under the law of motion induced by $\mathcal{V}$. Formally, for any $t \geq 0$,*

$$W_1\big(\text{Dist}(\Gamma_t \mid \Gamma_0 \sim \psi), \psi_{\mathcal{V}}\big) \leq \rho^t\, W_1(\psi, \psi_{\mathcal{V}}).$$

4.5 Key Chain of Inequalities

We are now able to build up a chain of inequalities linking the sampled residual and the true global error in $\mathcal{V}_\theta$: the sampled loss bounds the loss under[17] ψ_θ, the loss under ψ_θ bounds the *error* under ψ_θ, and the error under ψ_θ bounds the error under $\psi^\star$.

Definition 2 (Sampled loss)**.** Let $\{\Gamma_t\}_{t\geq 0}$, be a trajectory under $\mathcal{V}_\theta$,

$$\Gamma_{t+1} = \Gamma_{\mathcal{V}_\theta}^{\text{next}}(\Gamma_t, Z_t),$$
$$Z_t \sim \mathcal{D} \text{ i.i.d.},$$

which we have directly simulated using $\mathcal{V}_\theta$. Assume that this has been conducted with burn-in, so that,

$$\Gamma_0 \sim \psi_\theta,$$

where ψ_θ exists by Lemma 4.

The **sampled loss** along the trajectory is,

$$\widehat{S}_T \equiv \frac{1}{T}\sum_{t=0}^{T-1} r_\theta^2(\Gamma_t).$$

Lemma 5 and Theorem 1 provide bounds on the probability of observing $\widehat{S}_T$ for a given value of $\|\varepsilon_\theta\|$, in a frequentist sense. Confidence intervals and standard errors are provided in Appendix X. The Bayesian interpretation is given in Appendix X. Appendix **??**, Lemma **??** and Theorem **??**, give results in the more general case that the starting point Γ_0 is arbitrary and only finitely many periods have been discarded for burn-in.

Lemma 5 (The sampled loss bounds the ergodic mean loss with arbitrarily high probability)**.** *Under Assumptions 2, 4, 5 and 7–10, fix $\theta \in \Theta$. For any $\lambda > 0$,*

$$P\left(\|r_\theta\|_{2,\psi_\theta}^2 > \widehat{S}_T + \lambda\right) \leq \exp\left(-\frac{(1-\rho)^2}{2}\frac{T\lambda^2}{(\|r_\theta\| L_r \Delta_{\mathbb{\Gamma}})^2}\right),$$
$$P\left(\widehat{S}_T > \|r_\theta\|_{2,\psi_\theta}^2 + \lambda\right) \leq \exp\left(-\frac{(1-\rho)^2}{2}\frac{T\lambda^2}{(\|r_\theta\| L_r \Delta_{\mathbb{\Gamma}})^2}\right),$$

where L_r is the Lipschitz constant of r_θ defined in Lemma 3.

Proof. Let r_θ^2 be the local loss, so that by Lemma 3,

$$\text{Lip}(r_\theta^2) \leq L \equiv 2\,\|r_\theta\|\, L_r,$$

and let $C \equiv L\Delta_{\mathbb{\Gamma}}$, so that the relevant bound is $\exp\!\left(-2T(1-\rho)^2\lambda^2/C^2\right)$. ($C = 0$ only in the degenerate case where r_θ is constant, so assume $C > 0$.)

Since $\Gamma_0 \sim \psi_\theta$ and ψ_θ is invariant, $\Gamma_t \sim \psi_\theta$ for every t, so that,

$$\mathbb{E}\left[\widehat{S}_T\right] = \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[r_\theta^2(\Gamma_t)\right] = \int r_\theta^2\,\mathrm{d}\psi_\theta = \|r_\theta\|_{2,\psi_\theta}^2,$$

i.e. $\widehat{S}_T$ is an unbiased estimator of $\|r_\theta\|_{2,\psi_\theta}^2$. It remains to bound $P(\mathbb{E}\left[\widehat{S}_T\right] - \widehat{S}_T > \lambda)$.

[17] By "the X under ψ," I mean the p-norm of X under the (ergodic) distribution ψ, as described in Section X.

We will proceed by constructing the Doob martingale of $\widehat{S}_T$, which we will denote M_t: the conditional expectation of $\widehat{S}_T$ given the first $t+1$ observations. We will then show that the revisions in M_t are small, using Lemma 4 to show that the impact of one observation on the loss j periods later is at most $\rho^j C$. Having bounded the revisions to M_t, we use the Azuma–Hoeffding inequality (Lemma 9) to provide the explicit bound on the deviation of $\widehat{S}_T$ from its expectation.

Define the Doob Martingale of $\widehat{S}_T$ as the time-t forecast of the full sampled loss over all T periods. For $0 \le t \le T-1$, let

$$M_t \equiv \mathbb{E}\big[\widehat{S}_T \,\big|\, \Gamma_0, \ldots, \Gamma_t\big],$$

and let $M_{-1} \equiv \mathbb{E}\left[\widehat{S}_T\right]$ (conditioning on $\Gamma_0, \ldots, \Gamma_{-1}$, the case $t = 0$ below, means the plain expectation). By the law of iterated expectations, $M_{t-1} = \mathbb{E}[M_t \mid \Gamma_0, \ldots, \Gamma_{t-1}]$, so $(M_t)_{t=-1}^{T-1}$ is indeed a martingale. Furthermore, observe that $M_{T-1} = \widehat{S}_T$ exactly.

For all $0 \le t \le T-1$, the *revisions* of M_t are,

$$D_t \equiv M_t - M_{t-1}.$$

Observe that the D_t satisfy,

$$\mathbb{E}[D_t \mid \Gamma_0, \ldots, \Gamma_{t-1}] = 0 \quad \text{and} \quad \widehat{S}_T - \mathbb{E}\left[\widehat{S}_T\right] = M_{T-1} - M_{-1} = \sum_{t=0}^{T-1} D_t.$$

The Azuma–Hoeffding inequality (Lemma 9; **??**) states that if, conditional on $\Gamma_0, \ldots, \Gamma_{t-1}$, each D_t lies in an interval of length at most c, then,

$$P\big(\sum_t D_t > \lambda\big) \le \exp\big(-2\lambda^2/(Tc^2)\big),$$
$$P\big(-\sum_t D_t > \lambda\big) \le \exp\big(-2\lambda^2/(Tc^2)\big).$$

It remains to show that each D_t indeed lies in an interval of length at most $c = C/(T(1-\rho))$. The lemma immediately follows.

We can write the M_t explicitly. For $0 \le t \le T-1$, denote the conditional expected loss at time t, conditional on time-0 information, by,

$$r^2_{\theta,t}(\Gamma) \equiv \mathbb{E}\big[r^2_\theta(\Gamma_t) \,\big|\, \Gamma_0 = \Gamma\big].$$

Since the trajectory is Markov and time-homogeneous, for any $\tilde{t} \ge t$,

$$\mathbb{E}\left[r^2_\theta(\Gamma_{\tilde{t}}) \mid \Gamma_0, \ldots, \Gamma_t\right] = r^2_{\theta,\tilde{t}-t}(\Gamma_t).$$

Thus,

$$M_t = \frac{1}{T}\left(\sum_{j=0}^{t-1} r^2_\theta(\Gamma_j) + w_t(\Gamma_t)\right),$$

where,

$$w_t(\Gamma_t) \equiv \sum_{\tilde{t}=0}^{T-1-t} r^2_{\theta,\tilde{t}}(\Gamma_t)$$

is the forecast of the loss accumulated over periods $t, \dots, T-1$ when the state at t is Γ_t.

Therefore,

$$D_t = M_t - \mathbb{E}\big[M_t \mid \Gamma_0, \dots, \Gamma_{t-1}\big] = \frac{1}{T}\Big(w_t(\Gamma_t) - \mathbb{E}\big[w_t(\Gamma_t) \mid \Gamma_0, \dots, \Gamma_{t-1}\big]\Big).$$

Conditional on $\Gamma_0, \dots, \Gamma_{t-1}$, the only remaining variation in D_t comes from the $\frac{w_t(\Gamma_t)}{T}$ term. That is, D_t lies in an interval of length at most,

$$\frac{\sup w_t - \inf w_t}{T} \le \mathrm{Lip}(w_t)\,\frac{\Delta_{\mathbb{\Gamma}}}{T}$$

Finally,

$$\begin{aligned}\big|r^2_{\theta,j}(\Gamma) - r^2_{\theta,j}(\widetilde{\Gamma})\big| &\le \mathrm{Lip}(r^2_\theta)\, W_1\big(\mathrm{Dist}(\Gamma_j \mid \Gamma_0 = \Gamma), \mathrm{Dist}(\Gamma_j \mid \Gamma_0 = \widetilde{\Gamma})\big) \\ &\qquad \text{(by Kantorovich–Rubinstein, Lemma 8)} \\ &\le L\rho^j\, d(\Gamma, \widetilde{\Gamma}), \qquad \text{(by Lemma 4)}\end{aligned}$$

so that

$$\mathrm{Lip}(w_t) \le \sum_{j=0}^{T-1-t} L\rho^j \le \frac{L}{1-\rho},$$

and D_t lies in an interval of length at most $L\Delta_{\mathbb{\Gamma}}/(T(1-\rho)) = C/(T(1-\rho))$, as required. □

The asymptotic consequence of Lemma 5 is meaningful in its own right.

Corollary 3 (The sample loss is a consistent estimator of the ergodic mean loss.)**.** *Under Assumptions 2, 4, 5 and 7–10, fix $\theta \in \Theta$. For every $\lambda > 0$ and $\delta > 0$ there exists a $\overline{T} \in \mathbb{N}$ such that, for any $T \ge \overline{T}$,*

$$P\left(\big|\widehat{S}_T - \|r_\theta\|^2_{2,\psi_\theta}\big| > \lambda\right) < \delta.$$

Lemma 6 (Under the approximator's ergodic distribution, the mean loss bounds the mean error)**.** *Under Assumptions 2–6, 9 and 10, for every $\theta \in \Theta$ and $p \in \{1, 2\}$,*

$$\|\varepsilon_\theta\|_{p,\psi_\theta} \le \frac{\|r_\theta\|_{p,\psi_\theta}}{1 - L_V L_G - \xi_{\max}}.$$

Proof. Let $Z_0 \sim \mathcal{D}$, independent of Γ_0, and,

$$\begin{aligned}\Gamma_0 &\sim \psi_\theta, \\ \Gamma_1 &\equiv \Gamma^{\text{next}}_{\mathcal{V}_\theta}(\Gamma_0, Z_0),\end{aligned}$$

so that $\Gamma_1 \sim \psi_\theta$ by invariance. By Lemma 2 at Γ_0, (requiring Assumption 6 for both sides to be non-negative),

$$(1 - L_V L_G)\,\varepsilon_\theta(\Gamma_0) \le r_\theta(\Gamma_0) + \mathbb{E}\big[\xi(\Gamma_1)\,\varepsilon_\theta(\Gamma_1) \mid \Gamma_0\big],$$

By definition, $\|\varepsilon_\theta\|_{p,\psi_\theta} = \mathbb{E}[\varepsilon_\theta(\Gamma_0)^p]^{1/p}$, so that,

$$\begin{aligned}(1 - L_V L_G)\,\|\varepsilon_\theta\|_{p,\psi_\theta} &\le \mathbb{E}\Big[\Big(r_\theta(\Gamma_0) + \mathbb{E}\big[\xi(\Gamma_1)\,\varepsilon_\theta(\Gamma_1) \mid \Gamma_0\big]\Big)^p\Big]^{1/p} \\ &\le \|r_\theta\|_{p,\psi_\theta} + \mathbb{E}\Big[\mathbb{E}\big[\xi(\Gamma_1)\,\varepsilon_\theta(\Gamma_1) \mid \Gamma_0\big]^p\Big]^{1/p} \qquad \text{(by Minkowski's inequality)} \\ &\le \|r_\theta\|_{p,\psi_\theta} + \mathbb{E}\big[\xi(\Gamma_1)^p\,\varepsilon_\theta(\Gamma_1)^p\big]^{1/p} \qquad \text{(by Jensen's inequality)} \\ &\le \|r_\theta\|_{p,\psi_\theta} + \xi_{\max}\,\|\varepsilon_\theta\|_{p,\psi_\theta}. \qquad \text{(by Assumption 4.)}\end{aligned}$$

□

Lemma 7 (Distances between ergodic distributions and ergodic means are bounded)**.** *Under Assumptions 2, 5, 9 and 10, for every $\mathcal{V}_\theta \in \mathcal{N}$, the Wasserstein distance between the approximator's ergodic distribution ψ_θ and the true ergodic distribution $\psi^\star$ is bounded by a multiple of the true error, integrated over ψ_θ,*

$$W_1(\psi_\theta, \psi^\star) \leq \frac{L_G}{1-\rho} \, \|\varepsilon_\theta\|_{1,\psi_\theta} \,. \tag{7a}$$

Furthermore, the true ergodic mean error, integrated over $\psi^\star$, is bounded by the mean error over the approximator's distribution ψ_θ,

$$\|\varepsilon_\theta\|_{1,\psi^\star} \leq \left(1 + \frac{2L_{\mathcal{N}} L_G}{1-\rho}\right) \|\varepsilon_\theta\|_{1,\psi_\theta} \,. \tag{7b}$$

Proof. Let $Z_0 \sim \mathcal{D}$, independent of Γ_0 and $\widetilde{\Gamma}_0$, and,

$$\begin{aligned} \Gamma_0 &\sim \psi_\theta, & \widetilde{\Gamma}_0 &\sim \psi^\star, \\ \Gamma_\theta^{\text{next}} &\equiv \Gamma_{\mathcal{V}_\theta}^{\text{next}}(\Gamma_0, Z_0), & \widetilde{\Gamma}_\star^{\text{next}} &\equiv \Gamma_{\mathcal{V}^\star}^{\text{next}}(\widetilde{\Gamma}_0, Z_0), \\ \Gamma_\star^{\text{next}} &\equiv \Gamma_{\mathcal{V}^\star}^{\text{next}}(\Gamma_0, Z_0). \end{aligned}$$

By invariance, $\Gamma_\theta^{\text{next}} \sim \psi_\theta$ and $\widetilde{\Gamma}_\star^{\text{next}} \sim \psi^\star$, so that,

$$\begin{aligned} W_1(\psi_\theta, \psi^\star) &= W_1\big(\text{Dist}(\Gamma_\theta^{\text{next}}), \text{Dist}(\widetilde{\Gamma}_\star^{\text{next}})\big) \\ &\leq W_1\big(\text{Dist}(\Gamma_\theta^{\text{next}}), \text{Dist}(\Gamma_\star^{\text{next}})\big) + W_1\big(\text{Dist}(\Gamma_\star^{\text{next}}), \text{Dist}(\widetilde{\Gamma}_\star^{\text{next}})\big) \quad \text{(by triangle inequality)} \\ &\leq W_1\big(\text{Dist}(\Gamma_\theta^{\text{next}}), \text{Dist}(\Gamma_\star^{\text{next}})\big) + \rho \, W_1(\psi_\theta, \psi^\star) \quad \text{(by Lemma 4)} \end{aligned}$$

Since the Wasserstein-1 distance is an infimum over couplings, and $(\Gamma_\theta^{\text{next}}, \Gamma_\star^{\text{next}})$ is a coupling,

$$\begin{aligned} W_1\big(\text{Dist}(\Gamma_\theta^{\text{next}}), \text{Dist}(\Gamma_\star^{\text{next}})\big) &\leq \mathbb{E}\, d\big(\Gamma^{\text{next}}(\Gamma_0, \mathcal{V}_\theta\Gamma_0, Z_0), \Gamma^{\text{next}}(\Gamma_0, \mathcal{V}^\star\Gamma_0, Z_0)\big) \\ &\leq L_G \, \mathbb{E}\, \|\mathcal{V}_\theta\Gamma_0 - \mathcal{V}^\star\Gamma_0\| \quad \text{by Assumption 5} \\ &= L_G \, \|\varepsilon_\theta\|_{1,\psi_\theta} \,. \end{aligned}$$

So that all together,

$$W_1(\psi_\theta, \psi^\star) \leq L_G \, \|\varepsilon_\theta\|_{1,\psi_\theta} + \rho \, W_1(\psi_\theta, \psi^\star),$$

yielding (7a) by rearrangement.

To derive (7b), note that by the triangle inequality and Assumption 9, for any $\Gamma, \widetilde{\Gamma} \in \mathbb{\Gamma}$,

$$\big|\varepsilon_\theta(\Gamma) - \varepsilon_\theta(\widetilde{\Gamma})\big| \leq \left\|\mathcal{V}_\theta\Gamma - \mathcal{V}_\theta\widetilde{\Gamma}\right\| + \left\|\mathcal{V}^\star\Gamma - \mathcal{V}^\star\widetilde{\Gamma}\right\| \leq 2L_{\mathcal{N}} \, d(\Gamma, \widetilde{\Gamma}).$$

Kantorovich–Rubinstein duality (Lemma 8) then provides,

$$\begin{aligned} \|\varepsilon_\theta\|_{1,\psi^\star} - \|\varepsilon_\theta\|_{1,\psi_\theta} &= \int \varepsilon_\theta \, \mathrm{d}\psi^\star - \int \varepsilon_\theta \, \mathrm{d}\psi_\theta \\ &\leq 2L_{\mathcal{N}} \, W_1(\psi_\theta, \psi^\star) \\ &\leq \frac{2L_{\mathcal{N}} L_G}{1-\rho} \, \|\varepsilon_\theta\|_{1,\psi_\theta} \quad \text{by (7a)}, \end{aligned}$$

yielding (7b) by rearrangement. □

4.6 Main Result

Theorem 1 (The sampled loss bounds the global error with arbitrarily high probability)**.** *Assume Assumptions 2–10 and fix $\theta \in \Theta$. Let $\widehat{S}_T$ be the sampled loss along the law of motion induced by $\mathcal{V}_\theta$. For any $\lambda > 0$,*

$$P\left(\|\varepsilon_\theta\|_{2,\psi_\theta} > \frac{\sqrt{\widehat{S}_T + \lambda}}{1 - L_V L_G - \xi_{\max}}\right) \leq \exp\left(-\frac{(1-\rho)^2}{2}\frac{T\lambda^2}{(\|r_\theta\| L_r \Delta_{\mathbb{\Gamma}})^2}\right),$$

$$P\left(\|\varepsilon_\theta\|_{1,\psi^\star} > \left(1 + \frac{2L_{\mathcal{N}} L_G}{1-\rho}\right)\frac{\sqrt{\widehat{S}_T + \lambda}}{1 - L_V L_G - \xi_{\max}}\right) \leq \exp\left(-\frac{(1-\rho)^2}{2}\frac{T\lambda^2}{(\|r_\theta\| L_r \Delta_{\mathbb{\Gamma}})^2}\right),$$

where L_r is the Lipschitz constant of r_θ defined in Lemma 3, $\xi_{\max} + L_V L_G < 1$ is the joint contraction index in Assumption 6, ρ is the mixing rate, $L_{\mathcal{N}}$ is the Lipschitz constant of the neural architecture, and $\widehat{S}_T$ is the mean sample loss over T periods.

Proof. Fix $\lambda > 0$. First,

$$\begin{aligned} P\left(\|\varepsilon_\theta\|_{2,\psi_\theta} > \frac{\sqrt{\widehat{S}_T + \lambda}}{1 - L_V L_G - \xi_{\max}}\right) &\leq P\left(\|r_\theta\|^2_{2,\psi_\theta} > \widehat{S}_T + \lambda\right) \quad \text{by Lemma 6 with } p = 2 \\ &\leq \exp\left(-\frac{(1-\rho)^2}{2}\frac{T\lambda^2}{(\|r_\theta\| L_r \Delta_{\mathbb{\Gamma}})^2}\right) \quad \text{by Lemma 5,} \end{aligned}$$

and,

$$\begin{aligned} \|\varepsilon_\theta\|_{1,\psi^\star} &\leq \left(1 + \frac{2L_{\mathcal{N}} L_G}{1-\rho}\right)\|\varepsilon_\theta\|_{1,\psi_\theta} \quad \text{by Lemma 7(b)} \\ &\leq \left(1 + \frac{2L_{\mathcal{N}} L_G}{1-\rho}\right)\|\varepsilon_\theta\|_{2,\psi_\theta} \quad \text{by Jensen's inequality.} \end{aligned}$$

□

5 Application: A Spatial Model with Savings

In this section, I study a spatial model with forward-looking households, consumption-savings, and aggregate weather shocks.

In order to demonstrate the sense in which nVFI can be used with different within-period solvers, I run the same nVFI outer loop, with the same neural network and the same training algorithm, for three different within-period solvers. Each within-period solver solves a version of the same model, but one is in continuous time, one is in discrete time and solved by grid search, and one is in discrete time and solved by EGM.

5.1 Model

Locations $i \in I$ are populated by a unit mass of infinitely-lived households who make forward-looking migration and consumption-savings decisions. Households differ in their location, their wealth and their income, and choose their location based on a set of idiosyncratic location preference shocks and bilateral moving costs. Households may save in a risk-free foreign-issued bond at a

fixed exogenous interest rate, subject to a no-borrowing limit. Each location possesses a perfectly-competitive firm which produces a perfectly-tradable numeraire good using a labour-only technology featuring external decreasing returns to scale. Each location has labour productivity shifter A_{it} which is stochastic and possibly correlated across space and time. For concreteness, let us imagine that this stochasticity results from the weather.

Space is discrete. Time is divided into periods t of one year each, and the weather changes only between periods. We write the model in two versions. In the continuous-time version, households consume and save continuously within the period, and switch income and move at any instant. In the discrete-time version, each of these happens once per period, in a fixed order.

However, in both the continuous and discrete time versions, the aggregate shock occurs only once per period and the effect of the weather is constant throughout the period. This is the convenient "shock ages" assumption originally used by XXGreaneycite.

I now first describe the primitives of the model which are common to the discrete and continuous time versions, then describe the differences.

5.1.1 Firms

At time t, a firm k in location i has production function,

$$y_{kit} = A_{it} \left(H_{it}\right)^{-\alpha} N_k,$$

where H_{it} is the total labor input in location i, equal to the population of location i, N_k is the labour input of firm k, and $\alpha \in (0, 1)$ is the degree of external decreasing returns to scale within locations.

The total population is normalized to $\sum_i H_{it} = |I|$, so that a location holding an even share of the population has $H_{it} = 1$.

The wage in location i is then,

$$w_{it} = A_{it} \left(H_{it}\right)^{-\alpha}, \tag{17}$$

per unit of time, and the total output is given by,

$$Y_{it} = A_{it} \, H_{it}^{1-\alpha}.$$

In this model, the external decreasing returns to scale within locations represents congestion net of any positive external returns to scale or agglomeration effects. Without it, the equilibrium in a steady state version of the model would be for all households to cluster in the location with the highest productivity. It also introduces the dependence of local wages on local populations.

5.1.2 Weather

Each period, let the vector of location-level productivities be drawn from a joint distribution. For simplicity, assume that the "weather" is a single state m_t, which exogenously impacts each location's productivity,

$$A_{it} = \bar{A}_i \, e^{\chi_i m_t},$$

and that the weather process is Markov,

$$m_{t+1} \sim \Omega(\cdot \mid m_t), \tag{18}$$

where $\bar{A}_i$ is the productivity of location i in normal weather, χ_i is the sensitivity of location i productivity to the weather, and Ω is the distribution of the weather state, conditional on the previous period's.

Furthermore, suppose that the aggregate weather state is discretized into a finite set of nodes.[18] In the general notation of Section 2.1, the weather is the portion of the aggregate state S not including the population, and its draw is the aggregate shock.

5.1.3 Households

At the beginning of each period t, each household h has three state variables: its wealth $a_{ht} \in [0, a^{\max}]$, its income state $z_{ht} \in \{z_L, z_H\}$ and its current location i_{ht}. A household with income state z in location i earns $w_{it}z$ per unit of time, and receives utility $\log(c)$ from consuming c. Households receive a fixed interest rate r on their savings, with $r < \rho$.

When a household has an opportunity to move, it observes a set of location preference shocks, $\{\varepsilon_{jh}\}_{j \in I}$, which are distributed i.i.d. from a Gumbel distribution with scale ψ^{-1}, and chooses its location j, paying a bilateral migration cost $\kappa_{i,j}$ in utility, equal to zero if $j = i$.

We write $x = (a, z, i)$ for the household's state, $X^{\text{start}} = X^{\text{end}} = [0, a^{\max}] \times \{z_L, z_H\} \times I$, and Λ_t^{start} for the household population distribution at the start of period t, so that the aggregate state at the start of the period is

$$\Gamma_t^{\text{start}} = \left(\Lambda_t^{\text{start}},\, m_t\right).$$

The location-level populations H_{it} are given by,

$$H_{it} = \sum_z \int_0^{a^{\max}} \Lambda_t^{\text{start}}(da, z, i).$$

I now turn to separately describing the features which are particular to the choice of discrete or continuous time.

5.1.4 Continuous-Time Version

Setup. Within each year t, let $\tau \in [0, 1]$ index time. The weather and the weather-impacted productivity shifters $\{A_{it}\}_{i \in I}$ are fixed within the period. All other state variables move continuously. A household with state $x = (a, z, i)$ earns income at rate $w_{it}(\tau)z$, consumes at rate c and saves the difference,

$$\dot{a} = ra + w_{it}(\tau)z - c.$$

Income shocks arrive at Poisson rate q. Moving opportunities arrive at Poisson rate λ. For each instant τ within year t, wages update continuously,

$$w_{it}(\tau) = A_{it}\left(|I| H_{it}(\tau)\right)^{-\alpha}.$$

HJB Equation. Fixing the aggregate state Γ_t^{start} at the beginning of the period, let

$$v(a, z, i, \tau) \equiv V(a, z, i, \tau\,;\, \Gamma_t^{\text{start}})$$

be the value at instant τ of period t of a household in state $x = (a, z, i)$. Under the assumption that the household's location preference shocks are i.i.d. Gumbel with scale ψ^{-1}, a household with a moving opportunity moves to j with probability

$$\pi_{ij}(a, z, \tau) = \frac{\exp\left(\psi\left(v(a, z, j, \tau) - \kappa_{i,j}\right)\right)}{\sum_k \exp\left(\psi\left(v(a, z, k, \tau) - \kappa_{i,k}\right)\right)},$$

[18] This does not preclude rich covariance structure of weather across locations. Where a simple grid over the entire joint space of weather realizations is infeasible, we only require some low-rank approximation or dynamic grid of joint weather realizations which can be used in place of the full continuous state space.

and the value satisfies the Hamilton–Jacobi–Bellman equation,[19]

$$\rho v = \partial_\tau v + \max_c \{\log c + \partial_a v\,(ra + w_{it}(\tau)z - c)\} + q\,(v(a,\tilde{z},i,\tau) - v) + \lambda\left(\frac{1}{\psi}\log\sum_j \exp\left(\psi\left(v(a,z,j,\tau) - \kappa_{i,j}\right)\right) - v\right), \quad (19)$$

where $v \equiv v(a,z,i,\tau)$. At the borrowing limit, the state constraint is,

$$\partial_a v(0,z,i,\tau) \geq 1/\left(w_{it}(\tau)z\right).$$

At the final instant of the period, $\tau = 1$,

$$V^{\text{end}}(\cdot\,;\,\Gamma_t^{\text{start}}) \equiv v(\cdot,1). \quad (20)$$

Kolomogorov forward equation. Let $g(a,z,i,\tau)$ describe the population at instant τ, with $g(\cdot,0)$ describing Λ_t^{start}. Because $r < \rho$, households whose income falls short of the consumption they would like run their wealth down to the borrowing limit and stay there, so g is a density in wealth together with a mass at the borrowing limit. Away from the limit, it follows the Kolmogorov forward equation,

$$\partial_\tau g = -\partial_a\left[\left(ra + w_{it}(\tau)z - c(a,z,i,\tau)\right)\,g\right] + q\,(g(a,\tilde{z},i,\tau) - g) + \lambda\left(\sum_j \pi_{ji}(a,z,\tau)\,g(a,z,j,\tau) - g\right), \quad (21)$$

where $c(a,z,i,\tau)$ is the solution to Equation 19.

The population at the end of the period is,

$$\Lambda_t^{\text{end}} \equiv g(\cdot,1)$$

Shock ages. Between periods, the weather draws $m_{t+1} \sim \Omega(\cdot \mid m_t)$, and $\Gamma_{t+1}^{\text{start}} = (\Lambda_t^{\text{end}}, m_{t+1})$. The end-of-period value function is the expected value at the start of the next period,

$$V^{\text{end}}(x;\,\Gamma_t^{\text{start}}) = \sum_{m^{\text{next}}} \Omega(m^{\text{next}} \mid m_t)\,V^{\text{start}}\left(x;\,(\Lambda_t^{\text{end}}, m^{\text{next}})\right), \quad (22)$$

with $V^{\text{start}}(\cdot\,;\,\Gamma_t^{\text{start}}) = v(\cdot,0)$. Discounting occurs within the period, through ρ, not instantaneously at the shock age.

Note that the within-period problem, although deterministic, does represent a fixed point problem of its own, which for fine enough computational increments of τ will require an inner loop to solve. Wages depend on the population path $\{H_{it}(\tau)\}_{\tau\in[0,1]}$, and the path depends on the wages. However, due to the absence of aggregate shocks within the period, the within-period problem is nonetheless well-defined and deterministic.

[19] Dropping the Euler-Mascheroni constant as is standard.

5.1.5 Discrete-Time Version

In discrete time, simply has one opportunity to move per period, in the "location decision" stage of the household's problem. Over the course of the period, the household:

1. Observes the set of location-level productivities $\{A_{jt}\}_{j \in I}$.
2. Draws its income state for the period, $z_{h,t+1}$, from $P(\cdot \mid z_{ht})$.
3. Collects its interest and its location-specific income, with resulting cash-on-hand $b_{ht} = Ra_{ht} + w_{it} z_{h,t+1}$.
4. Observes a set of location preference shocks, $\{\varepsilon_{jht}\}_{j \in I}$, which are distributed i.i.d. from a Gumbel distribution with scale ψ^{-1}.
5. Chooses its location for the following period, $i_{h,t+1}$, paying a bilateral migration cost $\kappa_{i_{ht}, i_{h,t+1}}$, equal to zero if $i_{h,t+1} = i_{ht}$.
6. Consumes c_{ht}, receiving utility $\log(c_{ht})$, and saves the rest, $a_{h,t+1} = b_{ht} - c_{ht} \geq 0$.

To set the gross interest rate R and the transition probability for income P to the values equivalent to the continuous r and Q in the continuous-time model, we set,

$$R \equiv e^r \quad \text{and} \quad P \equiv e^Q$$

In choosing its consumption and next location, the household maximizes its present-discounted total future utility,

$$\max \mathbb{E}_t \left[\sum_{s=t}^{\infty} \beta^{s-t} \left(\log(c_{hs}) - \kappa_{i_{hs}, i_{h,s+1}} + \varepsilon_{i_{h,s+1} hs} \right) \right],$$

where to set value equivalent to the continuous-time model, we set $\beta = e^{-\rho}$.

Wages are set by the start-of-period population H_{it}. This is immaterial to the results but makes the solution substantially faster as it avoids the need for a within-period inner loop for wages. I describe this in further detail in Appendix X.

Under the assumption that the location preference shocks are i.i.d. Gumbel with scale ψ^{-1},

$$M(b, z, i; \Gamma_t^{\text{start}}) = \frac{1}{\psi} \log \sum_j \exp\left(\psi \left(W(b, z, j; \Gamma_t^{\text{start}}) - \kappa_{i,j} \right) \right),$$

and a household with cash on hand b and income state z moves from i to j with probability,

$$\pi_{ij}(b, z) = \frac{\exp\left(\psi \left(W(b, z, j; \Gamma_t^{\text{start}}) - \kappa_{i,j} \right) \right)}{\sum_k \exp\left(\psi \left(W(b, z, k; \Gamma_t^{\text{start}}) - \kappa_{i,k} \right) \right)}.$$

Furthermore, the law of motion for the household population is given by,

$$\Lambda_t^{\text{end}} \left(\mathcal{A} \times \{z^{\text{end}}\} \times \{j\} \right) = \sum_{i,z} \int P(z^{\text{end}} \mid z)\, \pi_{ij}(b, z^{\text{end}})\, \mathbf{1} \left\{ a^{\text{end}}(b, z^{\text{end}}, j) \in \mathcal{A} \right\} \Lambda_t^{\text{start}}(\, \mathrm{d}a, z, i), \tag{23}$$

for every set of wealth levels $\mathcal{A}$.

As in the continuous-time version, between periods, continuous time, $\Gamma_{t+1}^{\text{start}} = (\Lambda_t^{\text{end}}, m_{t+1})$ with $m_{t+1} \sim \Omega(\cdot \mid m_t)$.

The recursive formulation of the household's problem is,

$$V^{\text{start}}(a, z, i;\, \Gamma_t^{\text{start}}) = \sum_{z^{\text{end}}} P(z^{\text{end}} \mid z)\, M\left(Ra + w_{it} z^{\text{end}},\, z^{\text{end}},\, i;\, \Gamma_t^{\text{start}}\right), \tag{24}$$

$$M(b, z, i;\, \Gamma_t^{\text{start}}) = \mathbb{E}_{\varepsilon}\left[\max_j \varepsilon_j - \kappa_{i,j} + W(b, z, j;\, \Gamma_t^{\text{start}})\right], \tag{25}$$

$$W(b, z, j;\, \Gamma_t^{\text{start}}) = \max_{0 \le a^{\text{end}} < b,\ a^{\text{end}} \le a^{\max}} \log\left(b - a^{\text{end}}\right) + \beta\, V^{\text{end}}(a^{\text{end}}, z, j;\, \Gamma_t^{\text{start}}), \tag{26}$$

$$V^{\text{end}}(a^{\text{end}}, z, j;\, \Gamma_t^{\text{start}}) = \sum_{m^{\text{next}}} \Omega(m^{\text{next}} \mid m_t)\, V^{\text{start}}\left(a^{\text{end}}, z, j;\, (\Lambda_t^{\text{end}}, m^{\text{next}})\right). \tag{27}$$

5.2 Metrics

In order to evaluate the solutions, we must define a number of accuracy metrics.

Outer Bellman residual. This is the loss function which nVFI trains the network to minimize,

$$\text{SL} = \left(\frac{1}{T\,|X^{\text{end}}|} \sum_t \sum_{x^{\text{end}}} \left(\mathcal{V}_\theta \Gamma_t^{\text{start}}(x^{\text{end}}) - \widehat{v}_t^{\text{end}}(x^{\text{end}})\right)^2\right)^{1/2}.$$

This is reported in value units and as consumption equivalent, $e^{k\text{SL}} - 1$ with $k \in \{1-\beta, \rho\}$ for discrete and continuous time, respectively.

Population re-solve gap. In order to assess the extent to which the errors in the neural network affect the endogenous outcomes, we can look at the extent to which the neural network output and lookahead label produce different endogenous populations within the period.

That is, for each state Γ_t^{start}, we can solve for the resulting within-period populations by calling the within-period solver with $\mathcal{V}_\theta \Gamma_t^{\text{start}}$ or with $\widehat{v}_t^{\text{end}}$. The difference is,

$$G_t = \max_i \left|\frac{H_i^{\text{label}}}{H_i^{\theta}} - 1\right|.$$

Euler equation error. In discrete time, for each Γ_t^{start} and each household state $x = (b_k, z, j)$ at the consumption-savings stage, we can analytically compute the implied optimal consumption,

$$c^* = \min\left\{\left(\beta R \sum_{m^{\text{next}}} \Omega(m^{\text{next}} \mid m_t) \sum_{z^{\text{next}}} P(z^{\text{next}} \mid z) \sum_l \pi_{jl}\, c_{m^{\text{next}}}\left(b^{\text{next}}, z^{\text{next}}, l\right)^{-1}\right)^{-1},\ b_k\right\}.$$

In continuous time this is,

$$c^* = \left(\sum_{m^{\text{next}}} \Omega(m^{\text{next}} \mid m_t)\, D v_{m^{\text{next}}}^{\text{start}}\right)^{-1},$$

In both, we compute the population-weighted Euler equation error,

$$\text{EE}_{\text{DT}} = \sum_x \mu(x) \left|\frac{c^*(x) - c(x)}{c^*(x)}\right|.$$

Every accuracy metric is computed for one trajectory of 200 periods under the law of motion induced by a fixed neural network. The same initial state and sequence of weather shocks is used for all runs.The first 50 periods are discarded as burn-in, so the measured states are the last 200 periods out of a 250 period trajectory.

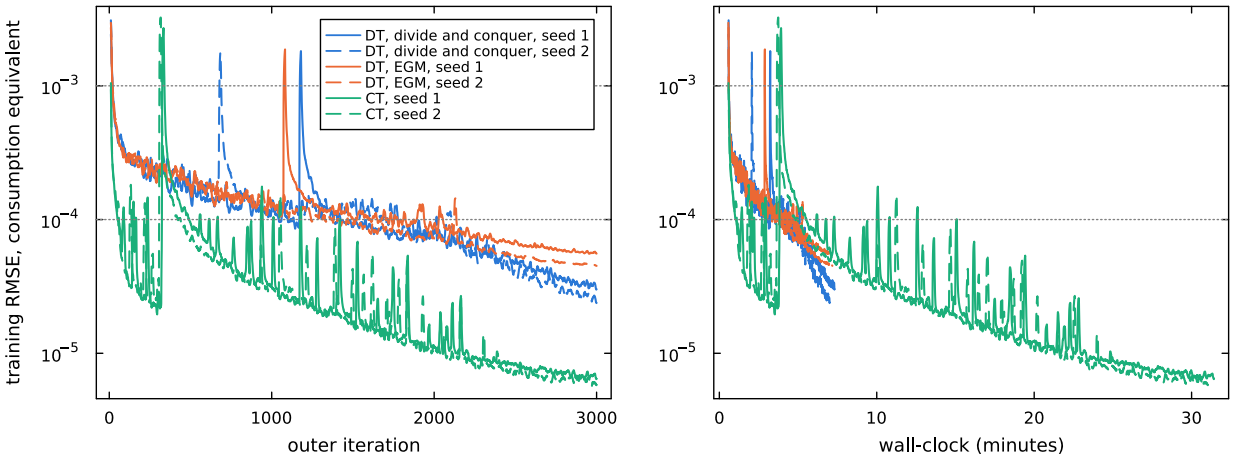


Figure 1: Loss curves for the networks for each of the discrete-time model with divide-and-conquer gridsearch policy solver, the discrete-time model with endogenous grid method solver, and the continuous-time model.

5.3 Results

Tables 1 and 2 report the accuracy and speed of the networks. Figure 1 displays their loss curves as a function of training iterations and training time.

One within-period solve of the discrete-time model takes 0.069 ms using the divide-and-conquer method and 0.089 ms using the endogenous gridpoints method. One within-period solve of the continuout-time model takes 8.7 ms. The gap between the outer iterations—0.12 and 0.12 s in discrete time, 0.61 s in continuous time—is much smaller, because each outer iteration includes exactly the same neural network training step.

Table 1: Accuracy of the networks. Each entry is the mean over two identical networks started at a different random seed. The error metrics are described in Section 5.2.

	Discrete time		Continuous time
	Divide and conquer	EGM	
Simulation loss	3.5×10^{-4}	8.9×10^{-4}	1.1×10^{-4}
	(3.8×10^{-4}, 3.1×10^{-4})	(9.5×10^{-4}, 8.2×10^{-4})	(1.1×10^{-4}, 1.0×10^{-4})
consumption equivalent	1.7×10^{-5}	4.3×10^{-5}	5.3×10^{-6}
	(1.9×10^{-5}, 1.5×10^{-5})	(4.6×10^{-5}, 4.0×10^{-5})	(5.6×10^{-6}, 5.1×10^{-6})
Re-solve gap, mean	2.2×10^{-4}	2.0×10^{-4}	4.7×10^{-5}
	(2.5×10^{-4}, 1.9×10^{-4})	(2.4×10^{-4}, 1.6×10^{-4})	(5.3×10^{-5}, 4.2×10^{-5})
Re-solve gap, maximum	1.0×10^{-3}	1.1×10^{-3}	3.6×10^{-4}
	(9.0×10^{-4}, 1.2×10^{-3})	(1.3×10^{-3}, 9.0×10^{-4})	(3.6×10^{-4}, 3.5×10^{-4})
Euler error, $\log_{10}$ mean	-2.7	-2.9	-3.6
	(-2.7, -2.7)	(-2.9, -2.9)	(-3.6, -3.7)
solver floor	-2.8	-3.3	$-\infty$
	(-2.8, -2.8)	(-3.3, -3.3)	($-\infty$, $-\infty$)
Euler error, $\log_{10}$ 99th pct.	-2.1	-2.1	-2.2
	(-2.1, -2.1)	(-2.1, -2.1)	(-2.2, -2.3)
solver floor	-2.1	-2.2	$-\infty$
	(-2.1, -2.1)	(-2.2, -2.2)	($-\infty$, $-\infty$)
Slope ratio, $\log_{10}$ mean	-2.9	-2.7	-3.6
	(-2.9, -2.9)	(-2.7, -2.7)	(-3.6, -3.7)
Slope ratio, $\log_{10}$ 99th pct.	-2.0	-2.0	-2.2
	(-2.0, -2.0)	(-2.0, -2.1)	(-2.2, -2.3)

Table 2: Speed of the three solvers. Each entry is the mean of two identical networks, started at different random seeds. The individual results are shown in parentheses below the mean value. Hardware: AMD Ryzen 9 7950X3D 16-Core CPU.

	Discrete time		Continuous time
	Divide and conquer	EGM	
Within-period solve (ms)	0.069	0.089	8.7
	(0.069, 0.069)	(0.089, 0.088)	(8.6, 8.9)
Fixed-point sweeps per solve	–	–	6.9
	(–, –)	(–, –)	(6.9, 6.9)
Label (ms)	0.52	0.60	38
	(0.49, 0.54)	(0.59, 0.61)	(38, 39)
Outer iteration (s)	0.12	0.12	0.61
	(0.12, 0.12)	(0.11, 0.12)	(0.61, 0.61)
Wall-clock to 10^{-3} (s)	36	36	36
	(36, 37)	(37, 36)	(36, 36)
Wall-clock to 10^{-4} (s)	190	210	65
	(180, 210)	(230, 200)	(64, 67)

6 Conclusion

A Standard Results

The following standard results are used in the proofs in Section 4.

Lemma 8 (Kantorovich–Rubinstein duality)**.** *For any two distributions μ and $\tilde{\mu}$ on a compact metric space, with W_1 the Wasserstein-1 metric of Section 4,*

$$W_1(\mu, \tilde{\mu}) = \sup\Big\{ \int f \, \mathrm{d}\mu - \int f \, \mathrm{d}\tilde{\mu} : \mathrm{Lip}(f) \le 1 \Big\}.$$

In particular, $\left| \int f \, \mathrm{d}\mu - \int f \, \mathrm{d}\tilde{\mu} \right| \le \mathrm{Lip}(f)\, W_1(\mu, \tilde{\mu})$ for every Lipschitz f. The result is due to **?***; see* **?***, Remark 6.5.*

Lemma 9 (Azuma–Hoeffding inequality)**.** *Let $(M_t)_{t=-1}^{T-1}$ be a martingale with respect to a filtration $(\mathcal{F}_t)_{t=-1}^{T-1}$, with increments $D_t \equiv M_t - M_{t-1}$, and suppose that for each $0 \le t \le T-1$ there are an $\mathcal{F}_{t-1}$-measurable A_t and a constant c_t with $A_t \le D_t \le A_t + c_t$. Then for every $\lambda > 0$,*

$$P\left(M_{T-1} - M_{-1} \ge \lambda\right) \le \exp\left(-\frac{2\lambda^2}{\sum_{t=0}^{T-1} c_t^2} \right),$$

and the same bound holds for $P\left(M_{-1} - M_{T-1} \ge \lambda\right)$. The inequality is due to **?** *and* **?***. This particular form appears in* **?***.*